# 1 Model Outputs:

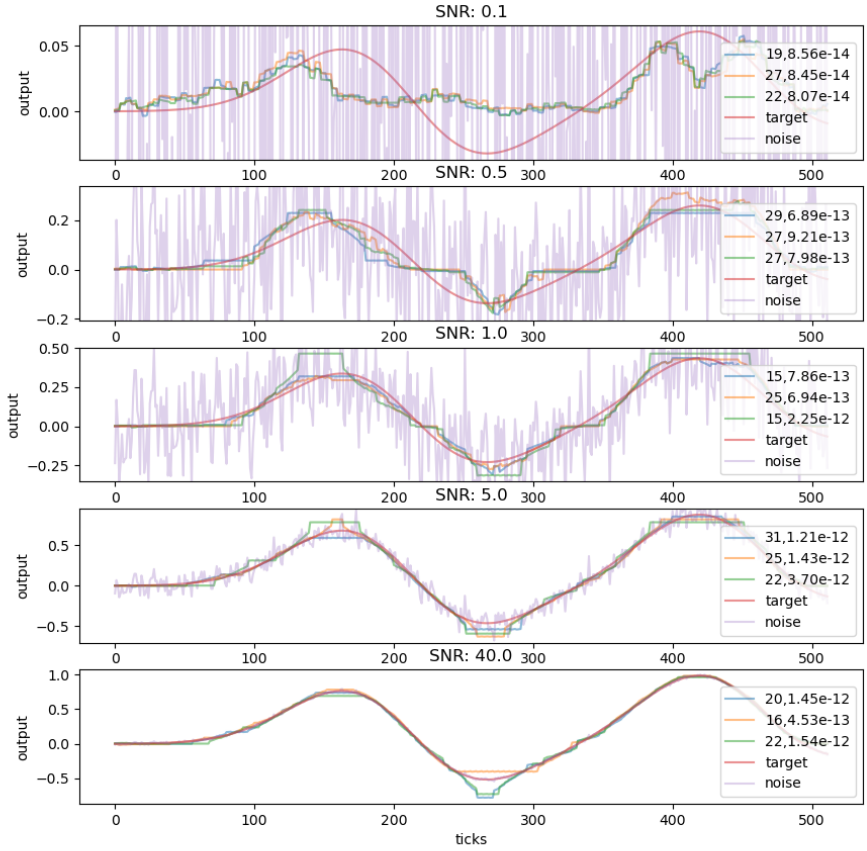



Figure 1: Outputs for the models for certain SNR datasets

Models were trained with synthetic datasets with an SNR around the specified value but with variation of a Gaussian with $\sigma = \frac{snr}{5*2}$. So 95% of the runs in each dataset are within 20% of the snr. In hindsight this is far too precise of datasets for this but for the goal of looking at the performance of n_codes on model behavior is more of a side note.

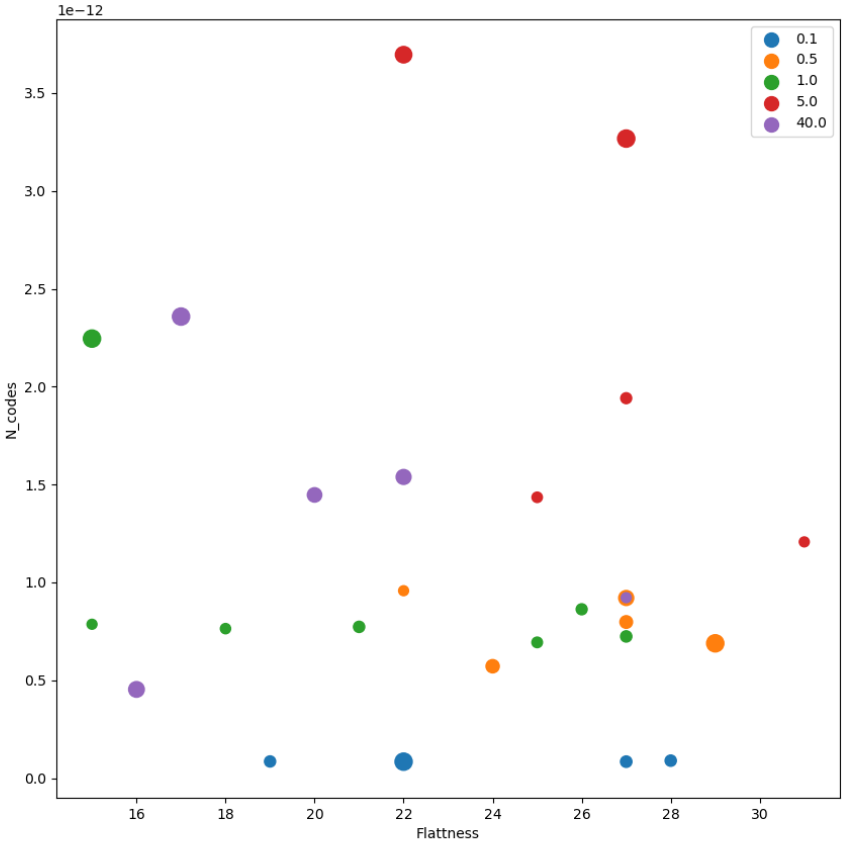
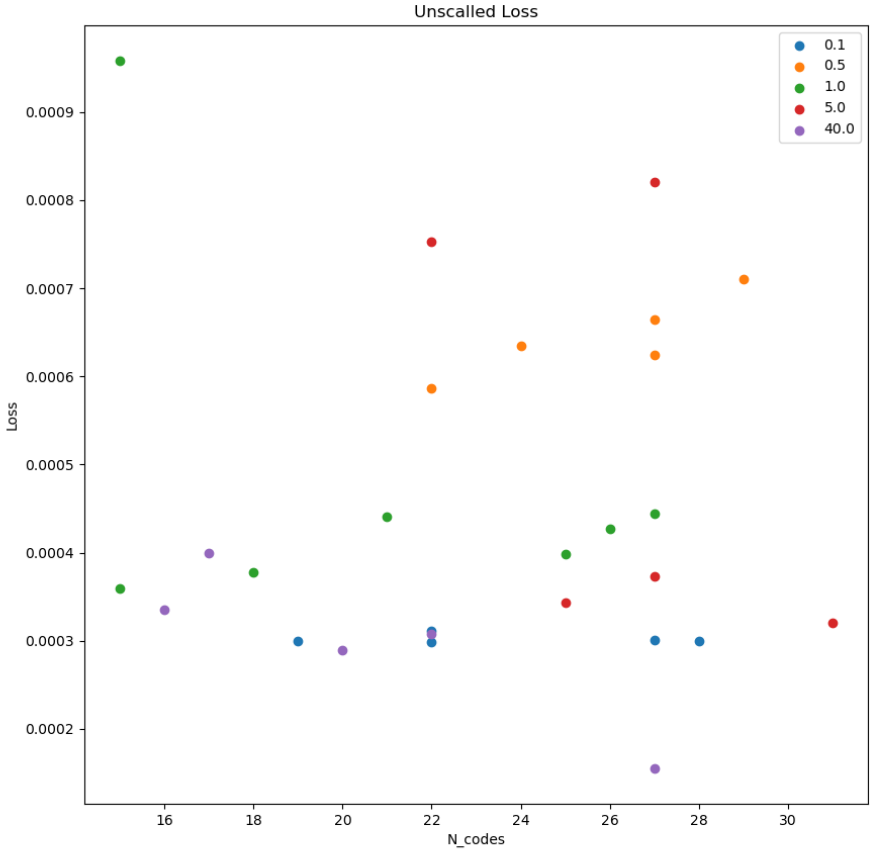


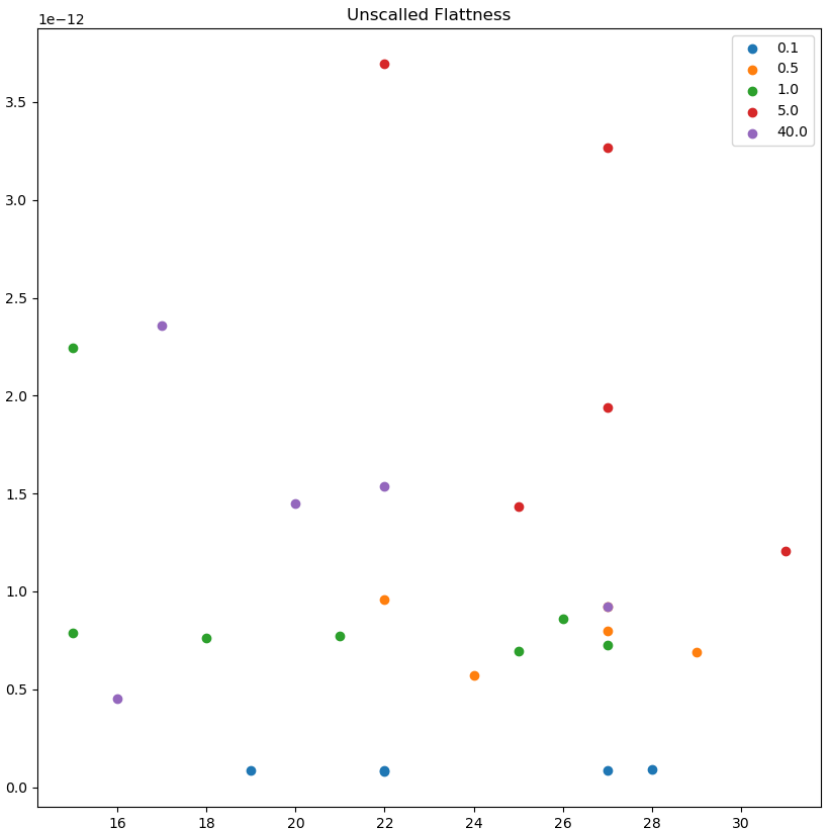

Figure 2: Comparison N_codes and the flattness metric with size denoting the relative loss

To measure the flatness seen in other runs a metric was used that should manage to track relative flatness of the outputs

$$flattness = \int (f(x))'' dx$$

with $f(x)$ being the data. The next two plots being the same but for the comparing relative to the loss instead.


Unscalled Loss
0.1
0.5
1.0
5.0
40.0
Loss
0.0009
0.0008
0.0007
0.0006
0.0005
0.0004
0.0003
0.0002
16
18
20
22
24
26
28
30
N_codes




While no apparent overall pattern exists across the datasets. For the $snr = 40$ dataset we see both flattness and loss drop with n_codes. For the lower snr runs we see either a decrease or an unchanging effect. The snr of 5 runs had no clear pattern and should be looked at again.